局部 NOETHERIAN 代数空间的完备化全忠实性

Counting scope. The primary problem is Tag 0GDR, marked below. All preceding numbered statements are supporting author context and are not counted as additional problems. This is a local pilot with a temporary ID, not production admission.

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Difficulty H2: 本题把完备化后的态射唯一代数化为 X 上的实际态射。证明在开补集和闭集完备化上先证唯一性，再反复选择 étale 仿射覆盖、代数化 rig-étale 态射并验证重叠相容以完成下降。仿射代数化定理允许引用，非仿射空间上的覆盖选择和全局粘合仍由本证明完成，判 H2。.

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Thus, we first fix a base scheme S. All rings, topological rings, schemes, algebraic spaces, and formal algebraic spaces and morphisms between these will be over S. Next, we fix a locally Noetherian algebraic space X and a closed subset $T \subset |X|$. We denote $U \subset X$ be the open subspace with $|U| = |X| \setminus T$. Picture

$$U \to X \quad |X| = |U| \amalg T$$

Given a morphism of algebraic spaces $f : X' \to X$, we will use the notation $U' = f^{-1}U$, $T' = |f|^{-1}(T)$, and $f_{/T} : X'_{/T'} \to X_{/T}$ as in Section section completion functor (Tag 0GDB). We will sometimes write $X'_{/T}$ in stead of $X'_{/T'}$ and more generally for a morphism $a : X' \to X''$ of algebraic spaces over X we will denote $a_{/T} : X'_{/T} \to X''_{/T}$ the induced morphism of formal algebraic spaces obtained by completing the morphism a along

Lemma 0.1. *Let S be a scheme. Let X, Y be locally Noetherian algebraic spaces over S. Let $T \subset |X|$ and $T' \subset |Y|$ be closed subsets. Let $a, b : X \to Y$ be morphisms of algebraic spaces over S such that $a|_{X\setminus T} = b|_{X\setminus T}$, such that $|a|(T) \subset T'$ and $|b|(T) \subset T'$, and such that $a_{/T} = b_{/T}$ as morphisms $X_{/T} \to Y_{/T'}$. Then $a = b$.*

Lemma 0.2. *Let $T \subset X$ be a closed subset of a Noetherian affine scheme X. Let W be a Noetherian affine formal algebraic space. Let $g : W \to X_{/T}$ be a rig-étale morphism. Then there exists an affine scheme X' and a finite type morphism $f : X' \to X$ étale over $X \setminus T$ such that there is an isomorphism $X'_{/f^{-1}T} \cong W$ compatible with $f_{/T}$ and g. Moreover, if $W \to X_{/T}$ is étale, then $X' \to X$ is étale.*

证明. The existence of X' is a restatement of Lemma lemma approximate by etale over complement (Tag 0AKG). The final statement follows from More on Morphisms, Lemma lemma check smoothness on infinitesimal nbhds (Tag 0A43). □

Lemma 0.3. *Assume we have*

(1) *Noetherian affine schemes X, X', and Y,*
(2) *a closed subset $T \subset |X|$,*
(3) *a morphism $f : X' \to X$ locally of finite type and étale over $X \setminus T$,*
(4) *a morphism $h : Y \to X$,*

(5) *a morphism $\alpha : Y_{/T} \to X'_{/T}$ over $X_{/T}$ (see proof for notation).*

Then there exists an étale morphism $b : Y' \to Y$ of affine schemes which induces an isomorphism $b_{/T} : Y'_{/T} \to Y_{/T}$ and a morphism $a : Y' \to X'$ over X such that $\alpha = a_{/T} \circ b_{/T}^{-1}$.

证明. The notation using the subscript $_{/T}$ in the statement refers to the construction which to a morphism of schemes $g : V \to X$ associates the morphism $g_{/T} : V_{/g^{-1}T} \to X_{/T}$ of formal algebraic spaces; it is a functor from the category of schemes over X to the category of formal algebraic spaces over $X_{/T}$, see Section section completion functor (Tag 0GDB). Having said this, the lemma is just a reformulation of Lemma lemma fully faithful etale over complement (Tag 0AKJ). □

Primary problem: Tag 0GDR

Lemma 0.4. *In the situation above. Let $f : X' \to X$ be a morphism of algebraic spaces which is locally of finite type and an isomorphism over U. Let $g : Y \to X$ be a morphism with Y locally Noetherian. Then completion defines a bijection*

$$\mathrm{Mor}_X(Y, X') \longrightarrow \mathrm{Mor}_{X_{/T}}(Y_{/T}, X'_{/T})$$

In particular, the functor (equation completion functor (Tag 0AR5)) is fully faithful.

证明. Let $a, b : Y \to X'$ be morphisms over X such that $a_{/T} = b_{/T}$. Then we see that a and b agree over the open subspace $g^{-1}U$ and after completion along $g^{-1}T$. Hence $a = b$ by Lemma 0.1. In other words, the completion map is always injective.

Let $\alpha : Y_{/T} \to X'_{/T}$ be a morphism of formal algebraic spaces over $X_{/T}$. We have to prove there exists a morphism $a : Y \to X'$ over X such that $\alpha = a_{/T}$. The proof proceeds by a standard but cumbersome reduction to the affine case and then applying Lemma 0.3.

Let $\{h_i : Y_i \to Y\}$ be an étale covering of algebraic spaces. If we can find for each i a morphism $a_i : Y_i \to X'$ over X whose completion $(a_i)_{/T} : (Y_i)_{/T} \to X'_{/T}$ is equal to $\alpha \circ (h_i)_{/T}$, then we get a morphism $a : Y \to X'$ with $\alpha = a_{/T}$. Namely, we first observe that $(a_i)_{/T} \circ \mathrm{pr}_1 = (a_j)_{/T} \circ \mathrm{pr}_2$ as morphisms $(Y_i \times_Y Y_j)_{/T} \to X'_{/T}$ by the agreement with α (this uses that completion $_{/T}$ commutes with fibre products). By the injectivity already proven this shows that $a_i \circ \mathrm{pr}_1 = a_j \circ \mathrm{pr}_2$ as morphisms $Y_i \times_Y Y_j \to X'$. Since X' is an fppf sheaf this means that the collection of morphisms a_i descends to a morphism $a : Y \to X'$. We have $\alpha = a_{/T}$ because $\{(a_i)_{/T} : (Y_i)_{/T} \to X'_{/T}\}$ is an étale covering.

By the result of the previous paragraph, to prove existence, we may assume that Y is affine and that $g : Y \to X$ factors as $g_1 : Y \to X_1$ and an étale morphism $X_1 \to X$ with X_1 affine. Then we can consider $T_1 \subset |X_1|$ the inverse image of T and we can set $X'_1 = X' \times_X X_1$ with projection $f_1 : X'_1 \to X_1$ and

$$\alpha_1 = (\alpha, (g_1)_{/T_1}) : Y_{/T_1} = Y_{/T} \longrightarrow X'_{/T} \times_{X_{/T}} (X_1)_{/T_1} = (X'_1)_{/T_1}$$

We conclude that it suffices to prove the existence for α_1 over X_1, in other words, we may replace $X, T, X', Y, f, g, \alpha$ by $X_1, T_1, X'_1, Y, g_1, \alpha_1$. This reduces us to the case described in the next paragraph.

Assume Y and X are affine. Recall that $(Y_{/T})_{red}$ is an affine scheme (isomorphic to the reduced induced scheme structure on $g^{-1}T \subset Y$, see Formal Spaces, Lemma lemma reduction completion (Tag 0GB9)). Hence $\alpha_{red} : (Y_{/T})_{red} \to (X'_{/T})_{red}$ has quasi-compact image E in $f^{-1}T$ (this is the underlying topological space of $(X'_{/T})_{red}$ by the same lemma as above). Thus we can find an affine

scheme V and an étale morpism $h : V \to X'$ such that the image of h contains E. Choose a solid cartesian diagram

$$\begin{array}{ccccc} Y'_{/T} & \dashrightarrow & W & \longrightarrow & V_{/T} \\ & \searrow & \downarrow & & \downarrow{\scriptstyle h_{/T}} \\ & & Y_{/T} & \xrightarrow{\alpha} & X'_{/T} \end{array}$$

By construction, the morphism $W \to Y_{/T}$ is representable by algebraic spaces, étale, and surjective (surjectivity can be seen by looking at the reductions, see Formal Spaces, Lemma lemma reduction surjective (Tag 0GB8)). By Lemma 0.2 we can write $W = Y'_{/T}$ for $Y' \to Y$ étale and Y' affine. This gives the dotted arrows in the diagram. Since $W \to Y_{/T}$ is surjective, we see that the image of $Y' \to Y$ contains $g^{-1}T$. Hence $\{Y' \to Y, Y \setminus g^{-1}T \to Y\}$ is an étale covering. As f is an isomorphism over U we have a (unique) morphism $Y \setminus g^{-1}T \to X'$ over X agreeing with α on completions (as the completion of $Y \setminus g^{-1}T$ is empty). Thus it suffices to prove the existence for Y' which reduces us to the case studied in the next paragraph.

By the result of the previous paragraph, we may assume that Y is affine and that α factors as $Y_{/T} \to V_{/T} \to X'_{/T}$ where V is an affine scheme étale over X'. We may still replace Y by the members of an affine étale covering. By Lemma 0.3 we may find an étale morphism $b : Y' \to Y$ of affine schemes which induces an isomorphism $b_{/T} : Y'_{/T} \to Y_{/T}$ and a morphism $c : Y' \to V$ such that $c_{/T} \circ b_{/T}^{-1}$ is the given morphism $Y_{/T} \to V_{/T}$. Setting $a' : Y' \to X'$ equal to the composition of c and $V \to X'$ we find that $a'_{/T} = \alpha \circ b_{/T}$, in other words, we have existence for Y' and $\alpha \circ b_{/T}$. Then we are done by replacing considering once more the étale covering $\{Y' \to Y, Y \setminus g^{-1}T \to Y\}$. $\square$